

0. Prerequisites

0.1. Differential and integral calculus of one and several variables

0.2. Complex numbers: $e^{i\theta} = \cos \theta + i \sin \theta$, etc

0.3. Taylor series: $f(x) = c_0 + c_1x + c_2x^2 + \dots \implies c_n = \frac{f^{(n)}(0)}{n!}$

0.4. Linear algebra, orthonormal bases: $\mathbf{v} = a_1\mathbf{e}_1 + \dots + a_n\mathbf{e}_n \implies a_i = \langle \mathbf{v}, \mathbf{e}_i \rangle$

0.5. Given the graph of a function, draw the graphs of its derivative and integral.

1. Fourier transform of functions $f : \mathbf{R} \rightarrow \mathbf{C}$

1.1 Definitions. The Fourier transform of an integrable function $f : \mathbf{R} \rightarrow \mathbf{C}$ is the function $\widehat{f} : \mathbf{R} \rightarrow \mathbf{C}$ defined by

$$\widehat{f}(y) := \int f(x)e^{-ixy}dx$$

and its inverse Fourier transform is defined by

$$\check{f}(y) := \frac{1}{2\pi} \int f(x)e^{ixy}dx$$

With some care, these definitions are extended to all square-integrable functions, not necessarily integrable. Then, the Fourier inversion theorem says that these two operations are inverses of each other. Thus, we have a representation of f as an (infinite) linear combination of sinusoidal functions, where the coefficients of the linear combination are given by $\widehat{f}(y)$:

$$f(x) = \frac{1}{2\pi} \int \widehat{f}(y)e^{ixy}dy$$

1.2. Energy conservation (Plancherel's theorem)

$$\int |f(x)|^2 dx = \frac{1}{2\pi} \int |\widehat{f}(y)|^2 dy$$

1.3. Convolution theorems

$$\widehat{f * g} = \widehat{f} \cdot \widehat{g} \qquad \widehat{fg} = \frac{1}{2\pi} \widehat{f} * \widehat{g}$$

Where the convolution of two functions is defined by

$$(f * g)(y) = \int f(x)g(y-x)dx$$

not to be confused with the correlation

$$(f \star g)(y) = \int \overline{f(x)}g(y+x)dx$$

1.4. General properties

f	$\widehat{f}$
real even	real even
real odd	imaginary odd
real	hermitian $\widehat{\widehat{f}}(y) = f(-y)$
$\lambda f + \mu g$	$\lambda \widehat{f} + \mu \widehat{g}$
$f(x/a)$	$ a \widehat{f}(ay)$
$f(x-a)$	$e^{-iay} \widehat{f}(y)$
$f'(x)$	$+iy \widehat{f}(y)$
$-ixf(x)$	$\widehat{f}'(y)$

1.5. Examples of transforms

$f(x)$	$\widehat{f}(y)$	$f(x)$	$\widehat{f}(y)$ (in the sense of distributions)
$\mathbf{1}_{[-a,a]}(x)$	$2a \operatorname{sinc}(ay)$	1	$2\pi\delta(y)$
$e^{-ax}\mathbf{1}_{[0,+\infty[}(x)$	$\frac{1}{a+iy}$	$\delta(x)$	1
$e^{-a x }$	$\frac{2a}{a^2+y^2}$	e^{iax}	$2\pi\delta(y-a)$
e^{-ax^2}	$\sqrt{\frac{\pi}{a}} e^{-\frac{y^2}{4a}}$	$\cos(ax)$	$\pi\delta(y-a) + \pi\delta(y+a)$
$\operatorname{sech}(ax)$	$\frac{\pi}{a} \operatorname{sech}\left(\frac{\pi}{2a}y\right)$	$\sin(ax)$	$-i\pi\delta(y-a) + i\pi\delta(y+a)$
		x^n	$2\pi i^n \delta^{(n)}(y)$
		$\sum_{n \in \mathbf{Z}} \delta(x-na)$	$\frac{2\pi}{a} \sum_{k \in \mathbf{Z}} \delta\left(y - \frac{2\pi k}{a}\right)$

2. Fourier series of functions $f : \mathbf{T} \rightarrow \mathbf{C}$

Let $\mathbf{T} = \mathbf{R}/2\pi\mathbf{Z}$ be the periodization of the interval $[0, 2\pi]$. The functions $f : \mathbf{T} \rightarrow \mathbf{C}$ are identified with the 2π -periodic functions on $\mathbf{R}$. They can be expressed as Fourier series, which is a linear combination of sinusoidal functions of integer frequencies:

$$f(\theta) = \sum_{n \in \mathbf{Z}} \widehat{f}(n) \exp^{in\theta} \quad (1)$$

the coefficients $\widehat{f}(n)$ are

$$\widehat{f}(n) = \frac{1}{2\pi} \int_0^{2\pi} f(\theta) e^{-in\theta} d\theta \quad (2)$$

(this formula follows by multiplying (1) with the function $e^{-in\theta}$ and integrating with respect to θ .)

Fourier series have analogous properties to Fourier integrals (see 1.4. above). Only the property for $f(x/a)$ has no direct equivalent, because it does not preserve the periodicity of the function. The energy conservation property $\int_0^{2\pi} |f|^2 = 2\pi \sum_{n \in \mathbf{Z}} |\widehat{f}(n)|^2$ can be used to compute the sums of many series.

3. Discrete Fourier transform of functions $f : \mathbf{P}_N \rightarrow \mathbf{C}$

Let $\mathbf{P}_N = \mathbf{Z}/N\mathbf{Z}$ be the integers modulo N . The functions $f : \mathbf{P}_N \rightarrow \mathbf{C}$ are the vectors of $\mathbf{C}^N$. The set of vectors

$$\mathbf{e}_n = \left(e^{\frac{2\pi i k n}{N}} \right)_{k=0, \dots, N-1}$$

for $n = 0, \dots, N-1$, is an orthogonal basis of $\mathbf{C}^N$, indeed $\mathbf{e}_p \cdot \mathbf{e}_q = N\delta_{pq}$.

The discrete Fourier transform (DFT) is the expression of $\mathbf{C}^N$ vectors in this basis. Thus, a vector $\mathbf{v} = (v_0, \dots, v_{N-1})$, can be expressed as

$$v_k = \sum_{n=0}^{N-1} \widehat{v}_n e^{\frac{2\pi i k n}{N}}$$

and the coefficients $\widehat{v}_n$ are recovered by computing $\mathbf{v} \cdot \mathbf{e}_n$:

$$\widehat{v}_n = \frac{1}{N} \sum_{k=0}^{N-1} v_k e^{-\frac{2\pi i k n}{N}}$$

The discrete Fourier transform has analogous properties to the other transforms. Here all the relationships are trivial to check using linear algebra (there are not convergence problems as in the previous cases).

4. Sampling

Consider a trigonometric polynomial of the form $f(\theta) = a_0 + a_1 e^{i\theta} + \dots + a_{N-1} e^{i(N-1)\theta}$. The coefficients a_k can be obtained from $f(\theta)$ by computing the Fourier series of the function $f : \mathbf{T} \rightarrow \mathbf{C}$, which is a finite sum.

Sampling theory provides another way to compute these coefficients. First, we evaluate the function f at N equally spaced points $\mathbf{v} := (f(2\pi k/N))_{k=0, \dots, N-1}$. Then, the vector of coefficients $\mathbf{a} = (a_0, \dots, a_{N-1})$ is the DFT of the vector $\mathbf{v}$.

There are similar relationships between the other transforms described above, describing how each transform commutes with sampling and interpolation operators.

5. Aliasing

Continuing with the sampling example above, suppose that $N = 2P + 1$ is odd, and notice that the two trigonometric polynomials

$$f(\theta) = a_0 + a_1 e^{i\theta} + \dots + a_{N-1} e^{i(N-1)\theta}$$

and

$$\tilde{f}(\theta) = a_0 + a_1 e^{i\theta} + \dots + a_P e^{iP\theta} + a_{P+1} e^{-iP\theta} + a_{P+2} e^{-i(P-1)\theta} \dots + a_{P+P} e^{-2i\theta} + a_{P+P+1} e^{-i\theta}$$

take exactly the same values at the points $\theta_k = \frac{2\pi k}{N}$ for $k = 0, \dots, N-1$. This happens because they have the same frequencies modulo N and the functions $e^{ik\theta}$ are 2π -periodic. The second choice, with frequencies symmetric around zero, is much more natural (since it is the only way to obtain real-valued signals!), and it is typically written as

$$f(\theta) = \sum_{k=-\lfloor N/2 \rfloor}^{\lfloor N/2 \rfloor - 1} a_k e^{ik\theta}$$

where the indices of the coefficients a_k are to be understood modulo N .

This is the simplest case of aliasing: different functions having the same samples. In the general case, we sample a function $f(\theta) = \sum_{n=0}^{N-1} a_n e^{in\theta}$ at M points $\theta_k = \frac{2\pi k}{M}$. The case $M = N$ is the perfect sampling rate (corresponding to the Shannon-Nyquist condition), and the coefficients a_n are the DFT of the samples.

The case $M > N$ is called *oversampling*, *zero-padding* or *zoom-in*, depending on the context. In that case we have

$$f(\theta_k) = \sum_{n=0}^{N-1} a_n e^{\frac{2\pi i k n}{M}} + \sum_{n=N}^{M-1} 0 \cdot e^{\frac{2\pi i k n}{M}} = \sum_{n=0}^{M-1} \text{ZP}(a)_n e^{\frac{2\pi i k n}{M}}$$

where $\text{ZP}(a_0, a_1, \dots, a_{N-1}) = (a_0, \dots, a_{N-1}, 0, \dots, 0)$ is the zero-padding of the vector a to length M .

The case $M < N$ is called *subsampling*, *aliasing*, *decimation* or *zoom-out*, depending on the context. In that case we have

$$f(\theta_k) = \sum_{n=0}^{N-1} a_n e^{\frac{2\pi i k (n \% M)}{M}} = \sum_{m=0}^{M-1} \left(\sum_{n \% M=m} a_n \right) e^{\frac{2\pi i k m}{M}} = \sum_{m=0}^{M-1} \text{AL}(a)_m e^{\frac{2\pi i k m}{M}}$$

where $\text{AL}(a)$ is the vector a *folded* to length M by summing the positions that are equal modulo M .

6. Fourier transform in several dimensions

If $f : \mathbf{R}^n \rightarrow \mathbf{C}$ is an integrable function, its Fourier transform is defined as:

$$\hat{f}(\mathbf{y}) = \int_{\mathbf{R}^n} f(\mathbf{x}) e^{-i\mathbf{x} \cdot \mathbf{y}} d\mathbf{x}$$

and the inverse transform is

$$\check{f}(\mathbf{y}) = \frac{1}{(2\pi)^n} \int_{\mathbf{R}^n} \hat{f}(\mathbf{x}) e^{i\mathbf{x} \cdot \mathbf{y}} d\mathbf{x}$$

Its properties are directly derived from those of the 1-dimensional Fourier transform by separability.

7. (Optional) Generalization: Pontryagin duality

The transforms described above are particular cases of a general construction called Pontryagin duality, which works on locally compact abelian groups G . On such a group, there is a natural measure called the *Haar measure* that allows to compute integrals of functions $f : G \rightarrow \mathbf{C}$ in a translation-invariant way. The *dual* of a group G is the set of morphisms $\mu : G \rightarrow \mathbf{C}^*$ (here $\mathbf{C}^*$ is the complex unit circle). The dual G^* is itself a group under the point-wise product of functions. Moreover, it is locally compact and abelian, and it has its own Haar integral. Finally, the Fourier transform of an integrable function $f : G \rightarrow \mathbf{C}$ is a function $\hat{f} : G^* \rightarrow \mathbf{C}$ defined as

$$\hat{f}(y) = \int_G f(x) \overline{y(x)} dx \quad (3)$$

And, likewise, given an integrable function $g : G^* \rightarrow \mathbf{C}$, its inverse transform is

$$\check{f}(x) = \int_{G^*} f(y) y(x) dy \quad (4)$$

And the Pontryagin duality theorem states that these two operations are inverses of each other. Notice that since $y(x)$ is a unit complex number, it is typically written as e^{iyx} .

On the table below we find the typical transforms as particular cases

	spatial domain	frequency domain	analysis	synthesis
general case	G	G^*	$\hat{f}(y) = \int_G f(x) e^{-iyx} dx$	$\check{f}(x) = \int_{G^*} f(y) e^{iyx} dy$
Fourier series	$\mathbf{T}$	$\mathbf{Z}$	$\hat{f}_n = \frac{1}{2\pi} \int_0^{2\pi} f(\theta) e^{-in\theta} d\theta$	$\check{f}(\theta) = \sum_{n \in \mathbf{Z}} \hat{f}_n e^{in\theta}$
Fourier transform	$\mathbf{R}$	$\mathbf{R}$	$\hat{f}(y) = \int_{\mathbf{R}} f(x) e^{-iyx} dx$	$\check{f}(x) = \frac{1}{2\pi} \int_{\mathbf{R}} f(y) e^{iyx} dy$
DFT	$\mathbf{Z}_N$	$\mathbf{Z}_N$	$\hat{f}_k = \frac{1}{N} \sum_{n=0}^{N-1} f_n e^{-ikn}$	$\check{f}_n = \sum_{k=0}^{N-1} \hat{f}_k e^{ikn}$
DTFT	$\mathbf{Z}$	$\mathbf{T}$	$\hat{f}(\theta) = \sum_{n \in \mathbf{Z}} f_n e^{-in\theta}$	$\check{f}_n = \frac{1}{2\pi} \int_0^{2\pi} \hat{f}(\theta) e^{in\theta} d\theta$

Notice that the placement of many multiplicative factors in this table seems arbitrary. Indeed the Haar measure is unique up to a multiplicative constant, and this leads to different conventions for the factors.